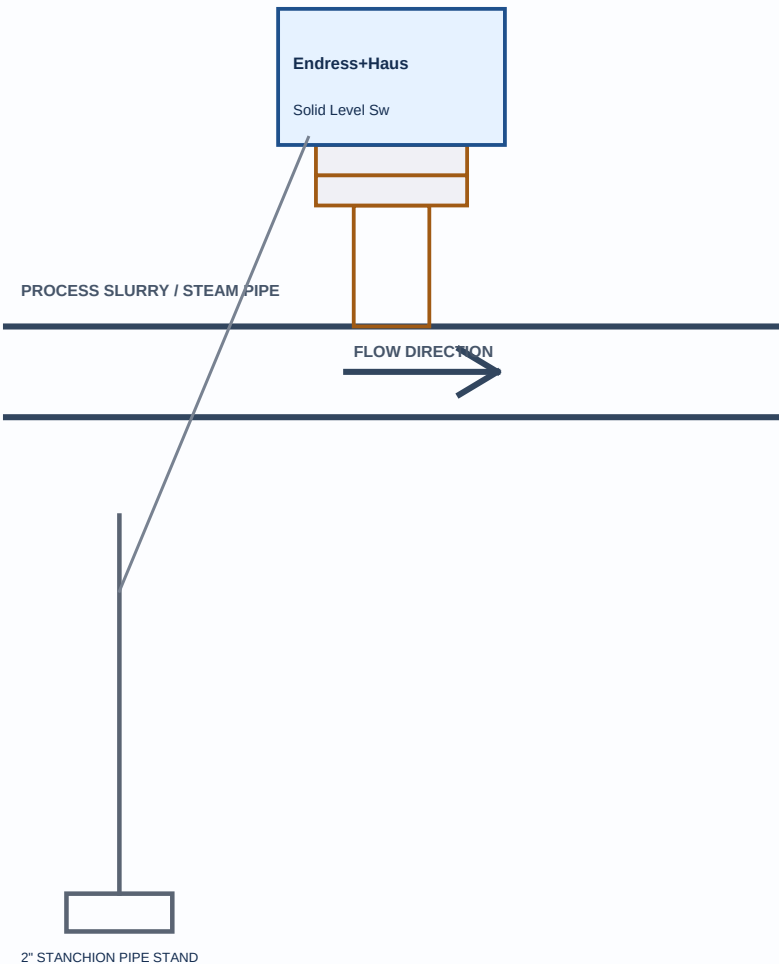


TEMPLATE TYP-13: SOLID LEVEL SWITCH

TYPICAL TAGS: LSH-60101 Series

SERVICE: Starch Powder Silo High Level Overfill Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

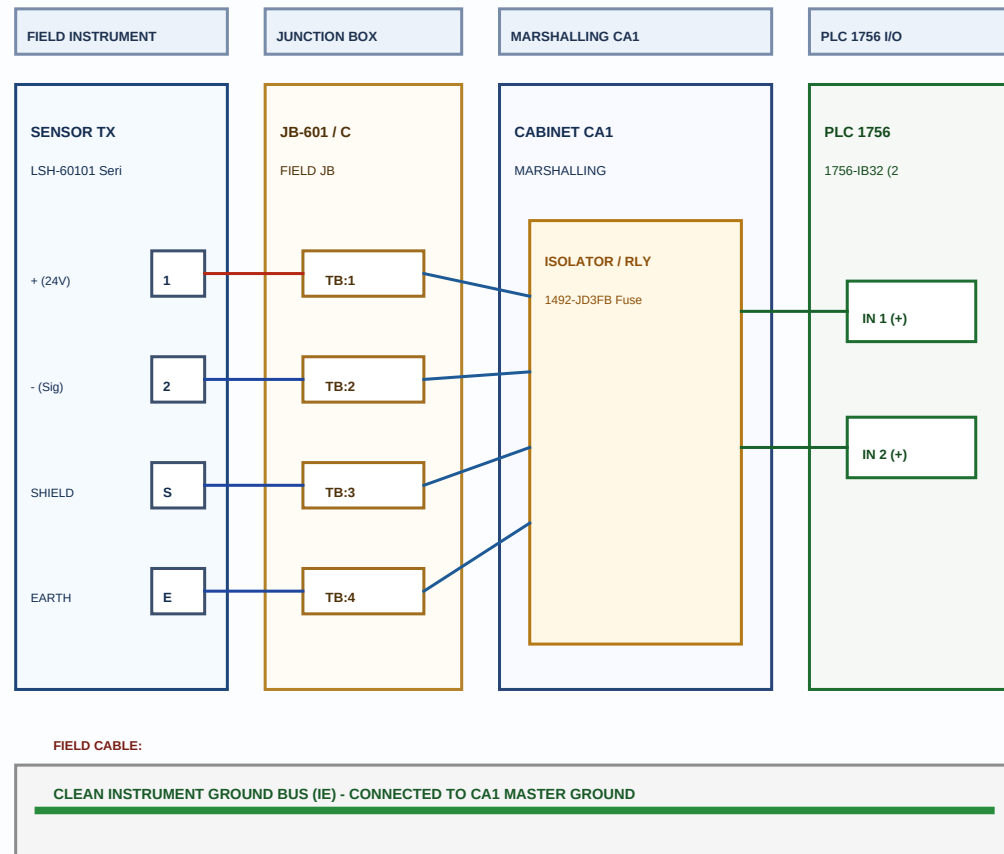
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1-1/2" NPT Male / DN50 Cl.15..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66 Cast Aluminum..
Hazardous Class:	ATEX II 1/2D Ex ta/tb IIIC T..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



FIELD CABLE:

CLEAN INSTRUMENT GROUND BUS (IE) - CONNECTED TO CA1 MASTER GROUND

ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: Silo Full Alarm Interlock
Power Supply:	24V DC (19-55V DC, 1.5W)
Field Cable Type:	4Cx1.5mm² Shielded Overall (Belden 9534)
Destination JB:	JB-601 / CA1 Marshalling
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Terminal + Relays
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Soliphant FTM51 (Vibrating Fork for Fine Bulk Solids)

SERVICE: Starch Powder Silo High Level Overfill Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Soliphant FTM51 Powder Level S E+H Fork Extension SS316L (300mm)		1 EA
2	Welding Flange DN50 PN16 / Cl. Carbon Steel / SS316 Flange		1 EA
3	Weather Protective Sun / Rain SS304 Protective Roof		1 EA
4	Dust-Ex Cable Gland M20x1.5 Nickel-plated Brass ATEX Dust GI		1 EA
5	Grounding Wire 6mm² Green/Yell Copper Earthing Strap		1 EA

ENGINEERING INSTALLATION NOTES

- Install angled downward 20° to 30° on silo side wall to allow powder to slide off times easily.
- Do not mount directly in the filling stream or impact zone of falling starch powder.
- Ensure housing earth terminal is connected to silo metal structure to dissipate electrostatic charge.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONSOLID LEVEL SWITCH (POWDER FORK) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-013
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
15 / 25